

EQUIFLOW QA ASSESSMENT

Before & After QA Test Case Report & Database Migration Summary

LIVE APPLICATION HOST: <https://tactive.onrender.com/>

Generated: 2026-08-13 | DB: MongoDB Atlas Cloud

1. Executive Summary

EquipFlow is an enterprise-grade Asset & Equipment Reservation Platform. This document captures the complete verification lifecycle, including the seamless migration from SQLite to MongoDB Atlas, the 19-point Vitest API Automation Test Suite execution results, the deliberate Red Run failure capture, and the AI Change-Loop log.

2. Architecture & Database Migration (Before vs After)

Metric / Component	BEFORE (SQLite Local)	AFTER (MongoDB Atlas Cloud)
Database Engine	better-sqlite3 (Local File)	MongoDB Atlas (Cloud Cluster)
Connection Driver	Synchronous SQLite WAL	Mongoose 8 async connection
Primary Keys	Auto-increment Integer (1, 2, 3)	ObjectId ("60c72b2f...")
Environment Config	data/equipflow.db local file	mongodb+srv://... in .env
Test Execution	19/19 Passed (279ms)	19/19 Passed (15.78s)
Data Scalability	Single-node filesystem bound	Cloud Distributed Cluster

3. Test Suite Verification Results (19/19 PASS)

Category	Test Scenario & Expectation	Status
1. Happy Path	GET /api/assets -> Seeded inventory list returned	PASS [HTTP 200]
1. Happy Path	GET /api/health -> Health status check	PASS [HTTP 200]
1. Happy Path	VIP User auto-confirmed reservation (No Admin queue)	PASS [HTTP 201]
1. Happy Path	Standard User reservation -> Enters PENDING queue	PASS [HTTP 201]
1. Happy Path	Admin can approve PENDING reservation	PASS [HTTP 200]
1. Happy Path	Admin can reject PENDING reservation	PASS [HTTP 200]
1. Happy Path	User can cancel own PENDING reservation	PASS [HTTP 200]
2. Business Rules	CONFLICT: Overlapping dates rejected	PASS [HTTP 409]
2. Business Rules	QUOTA: Standard user blocked at >3 active bookings	PASS [HTTP 400]
2. Business Rules	BLACKOUT: Booking blocked during admin maintenance window	PASS [HTTP 409]
2. Business Rules	PENALTY (VIP): 3 days overdue -> \$45 (50% VIP discount)	PASS [HTTP 200]
2. Business Rules	PENALTY (Standard): On-time return = \$0 penalty	PASS [HTTP 200]
3. Validation	REJECT: Inverted dates (start > end)	PASS [HTTP 409]
3. Validation	REJECT: Booking duration > 14 days limit	PASS [HTTP 409]
3. Validation	REJECT: Non-existent asset ID returns 404	PASS [HTTP 404]
3. Validation	REJECT: Unauthenticated request returns 401	PASS [HTTP 401]
3. Validation	REJECT: Standard user cannot create blackouts	PASS [HTTP 403]
3. Validation	REJECT: Missing required fields (Zod 400)	PASS [HTTP 400]
3. Validation	REJECT: Invalid date format (Zod 400)	PASS [HTTP 400]

FAIL tests/api/reservation.test.ts
x CONFLICT: Overlapping dates for same asset must be rejected (409)
AssertionError: expected 201 to be 409

4. Deliberate Red Run Verification

~~0-5/4 RED RUN EVIDENCE OUTPUT~~

As required by Stage 2 of the Tactive Assessment, a regression was deliberately injected into conflictEngine.ts to verify that the automated test suite catches regressions. The test suite correctly flagged the double-booking flaw with an assertion error (AssertionError: expected 201 to be 409).
Github Repository: <https://github.com/pskang1992/005-Tactive>
Live App Host: <https://tactive-frontend.com/>

5. AI Change Loop Summary

Submission Status: COMPLETE & READY FOR EVALUATION

Mid-development, the VIP Priority Tier feature was added (auto-approval + 50% late fee discount). The AI Change Loop caught stale test expectations, self-corrected the test assertions, and restored 100% test pass rate in 3 iterations.

